

Fundamentals of Solid State Physics

Superconductivity

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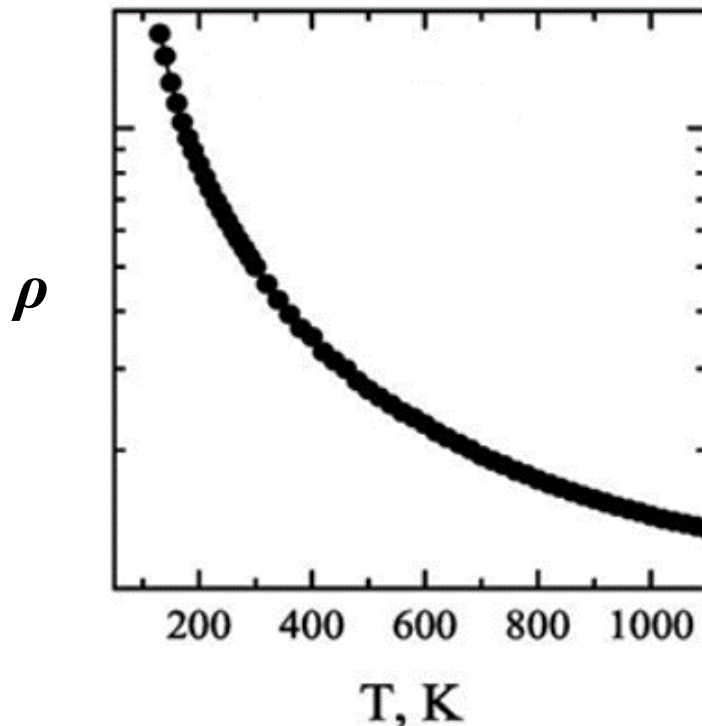
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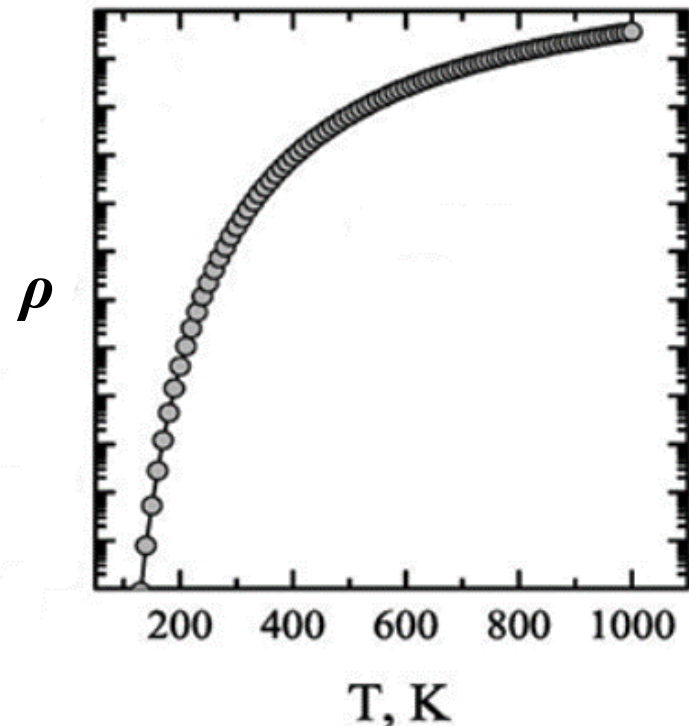
Resistivity ρ vs. Temperature

Metals and semiconductors have different temperature dependences of ρ

$$\sigma = ne\mu$$



Intrinsic Semiconductors



Metals

Resistivity ρ of Metals

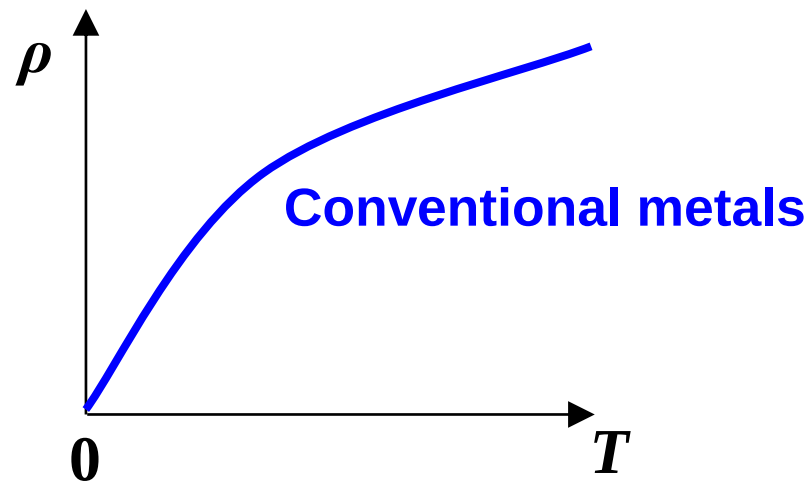
■ The Classical Model

- ▣ Resistivity ρ is always > 0 for metals, because of phonon scattering

$$\sigma = ne^2 \frac{\tau}{m}$$

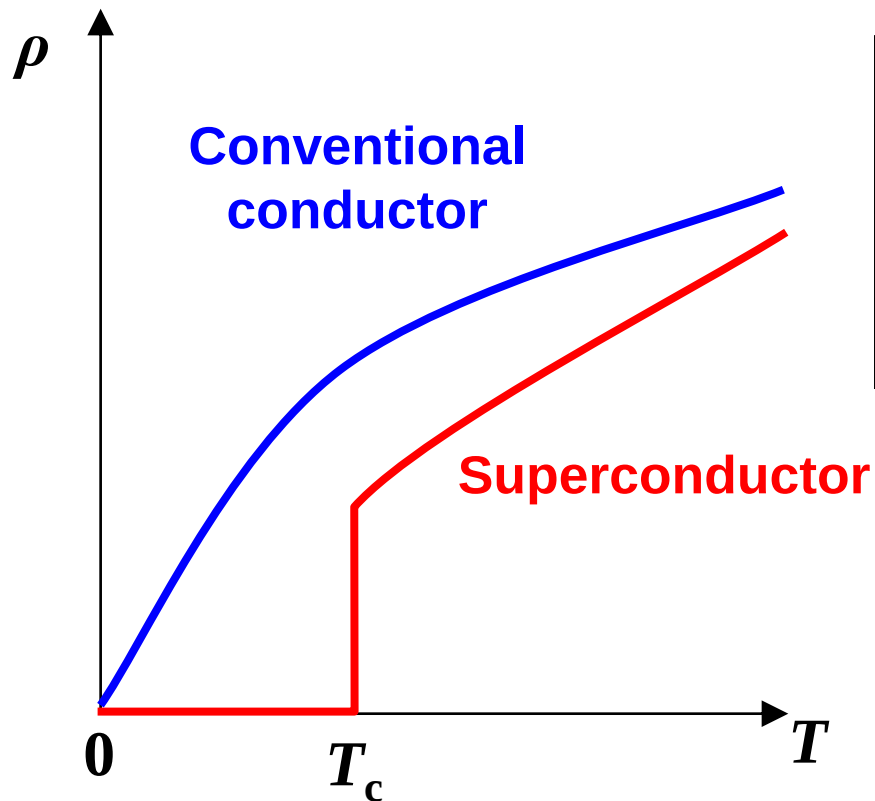
$$\rho = \frac{1}{\sigma}$$

when T decreases $\rightarrow \tau$ increases $\rightarrow \rho$ decreases

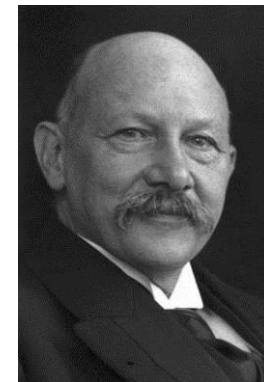


Superconductor 超导体

- Resistivity drops to 0 at transition temperature T_c
- Phonon scattering suddenly disappears



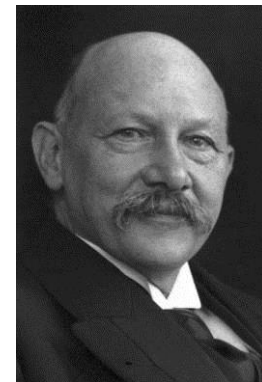
	T_c (K)
Hg	4.2
Al	1.1
Li	0.0004



H. Onnes (昂尼斯)
1913 Nobel Prize in Physics
for Low Temperature Physics

Superconductor 超导体

- Onnes's main focus is to get liquid helium ($T = 4\text{ K}$).
- Discovery of superconductivity is serendipity



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Nobel Prizes in Superconductivity

- **1913** **Low temperature physics**
- **1972** **BCS theory of superconductivity**
- **1973** **Tunneling effects in superconductors**
- **1987** **High temperature superconductors**
- **2003** **Theory of superconductors**
- **...**

Superconductor 超导体

KNOWN SUPERCONDUCTIVE ELEMENTS																		0																	
1A																2																			
1	H															He																			
IIA																																			
3	Li	4	Be															5	B	6	C	7	N	8	O	9	F	10	Ne						
11		12																13		14		15		16		17		18							
Na		Mg																Al		Si		P		S		Cl		Ar							
III B		IV B		VB		VIB		VIIB		VII		IB		IIB																					
19	K	20	Ca	21	Sc	22	Ti	23	Y	24	Cr	25	Mn	26	Fe	27	Co	28	Ni	29	Cu	30	Zn	31	Ga	32	Ge	33	As	34	Se	35	Br	36	Kr
37		38		39		40		41		42		43		44		45		46		47		48		49		50		51		52		53		54	
Rb		Sr		Y		Zr		Nb		Mo		Tc		Ru		Rh		Pd		Ag		Cd		In		Sn		Sb		Te		I		Xe	
55		56		57		72		73		74		75		76		77		78		79		80		81		82		83		84		85		86	
Cs		Ba		*La		Hf		Ta		W		Re		Os		Ir		Pt		Au		Hg		Tl		Pb		Bi		Po		At		Rn	
87		88		89		104		105		106		107		108		109		110		111		112													
Fr		Ra		+Ac		Rf		Ha		106		107		108		109		110		111		112													
																								SUPERCONDUCTORS.ORG											

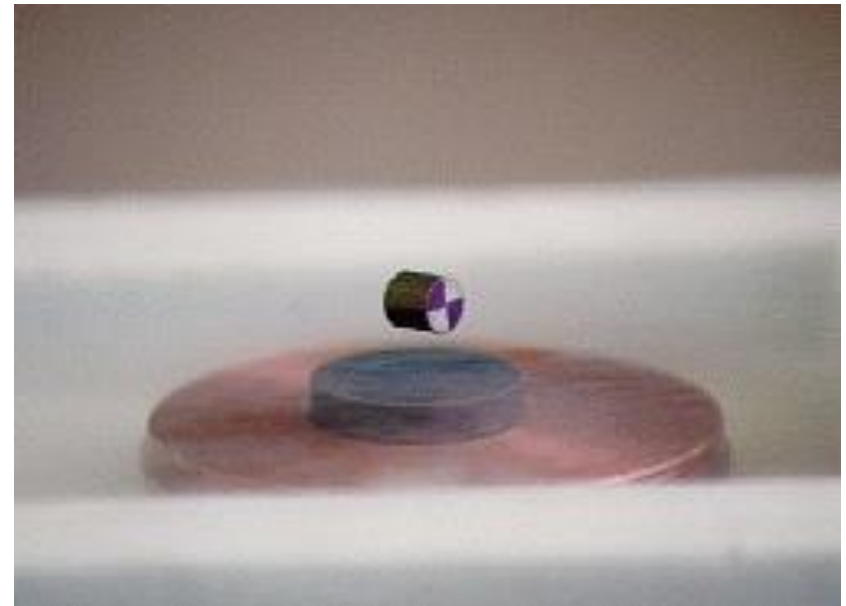
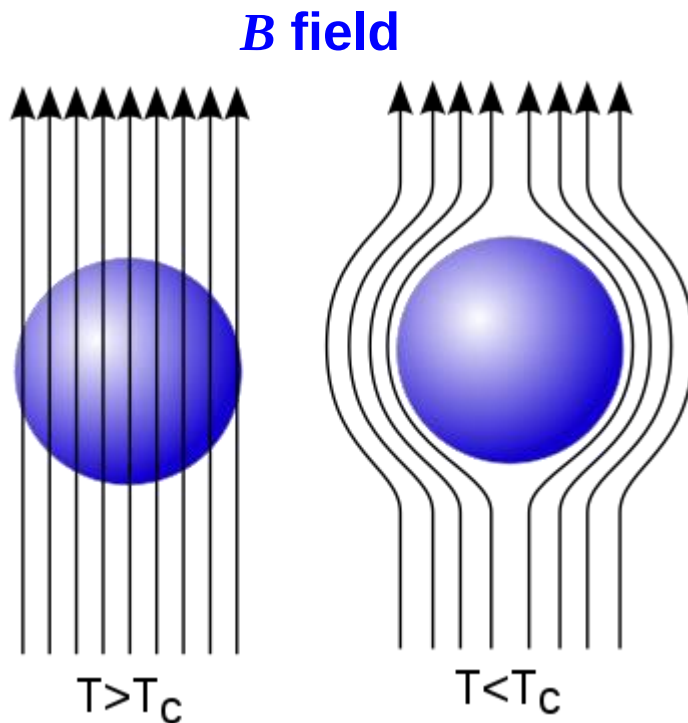
* Lanthanide Series

+ Actinide Series

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Superconductor 超导体

- Meissner effect 迈斯纳效应
 - ❑ Superconductors repel all the magnetic field inside
 - ❑ perfectly diamagnetic ($\chi = -1$)
 - ❑ Inside, $B = \mu_0\mu_r H = \mu_0(1+\chi)H = 0$



Superconductor 超导体

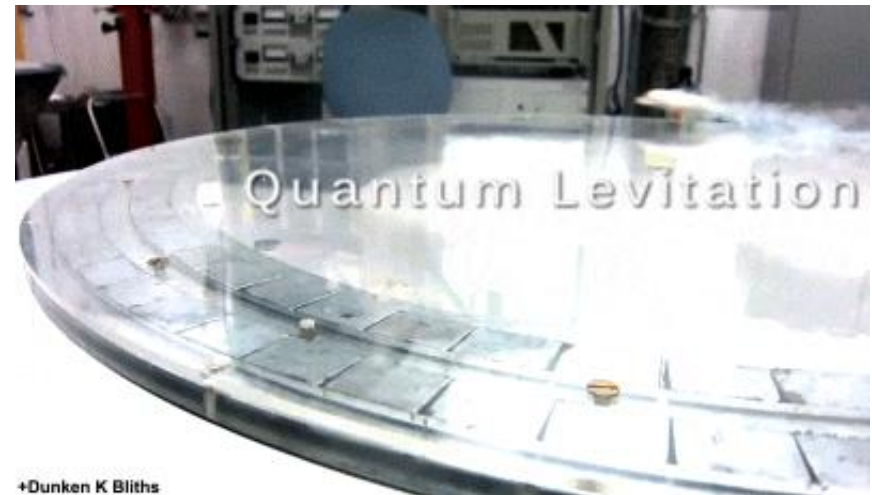
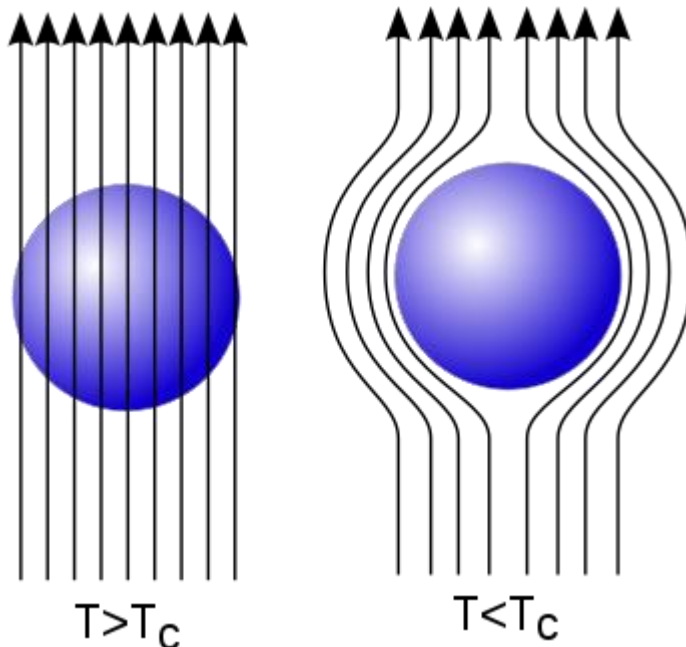
- **Meissner effect 迈斯纳效应**
 - Superconductors repel all the magnetic field inside
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- **A superconductor is not just a perfect conductor**
 - Meissner effect cannot be simply explained by $\rho = 0$
 - It can only be understood by quantum mechanics

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B field

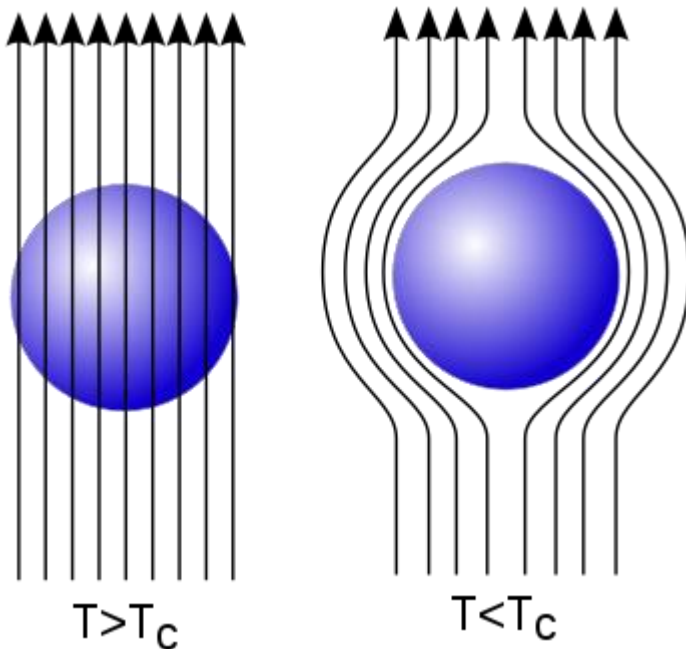


<https://wonderfuleengineering.com/these-15-magnet-gifs-will-show-you-the-power-of-magnetism/>

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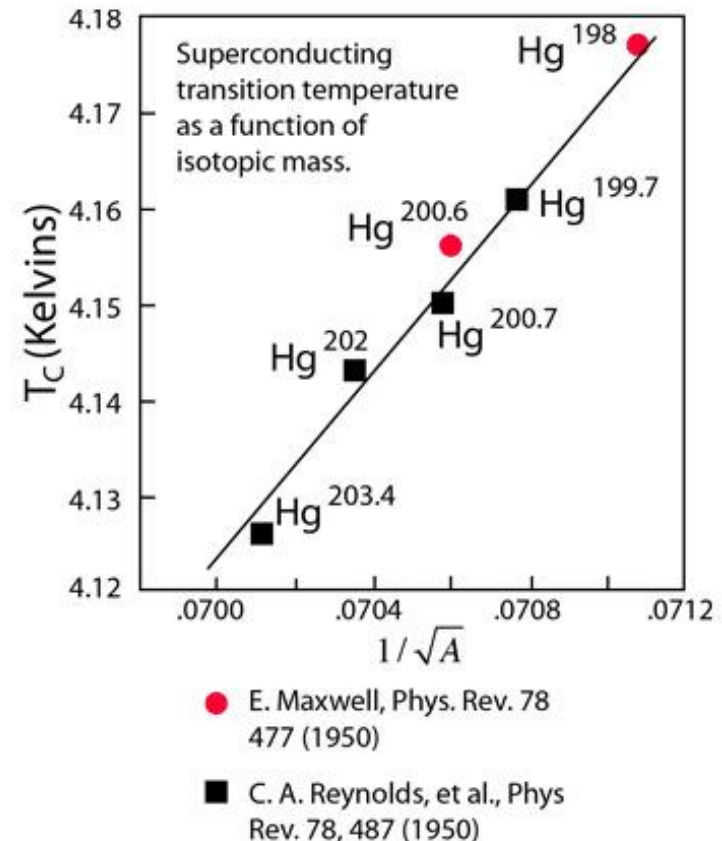
MagLev (磁悬浮列车)

Theory of Superconductors

- Some observations
 - Superconductors are usually not good conductors at room temperature
 - Good metals like Au, Ag, Cu cannot be superconductors
 - Critical temperature is depending on the atomic mass (Isotope Effect, 同位素效应)

- One conclusion

- Superconducting effect must be related to phonons!

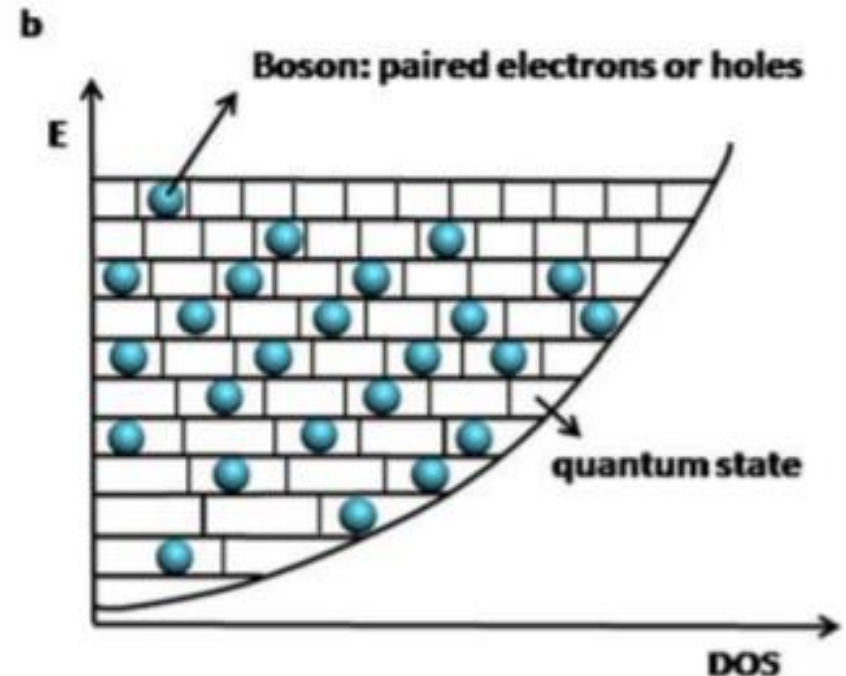
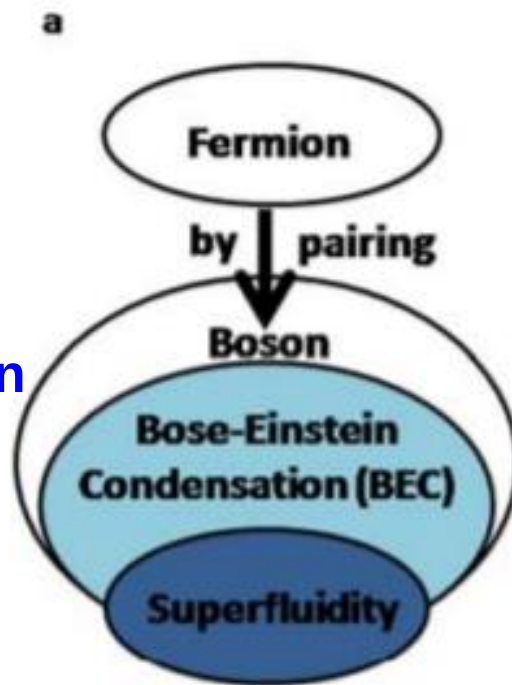


BCS Theory

- Pairs of electrons (Cooper's Pairs) move in the lattice coherently without phonon scattering

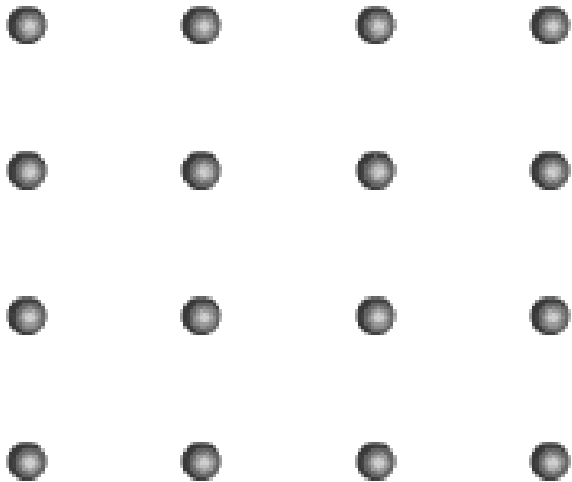
electron spin
 $= 1/2$ or $-1/2$
 $=$ Fermion

cooper pair spin
 $= 1$ or 0
 $=$ Boson



BCS Theory

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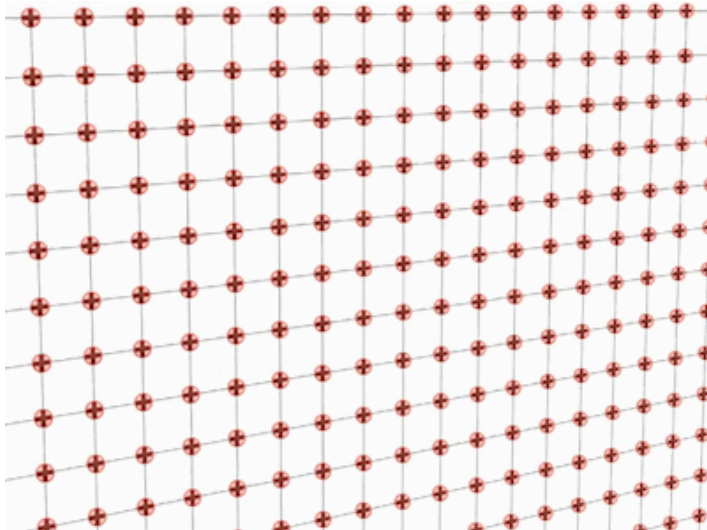
A Cooper Pair of electrons
moving in the lattice



Bardeen, Cooper and Schreffer
1972 Nobel Prize in Physics

BCS Theory

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**A Cooper Pair of electrons
moving in the lattice**

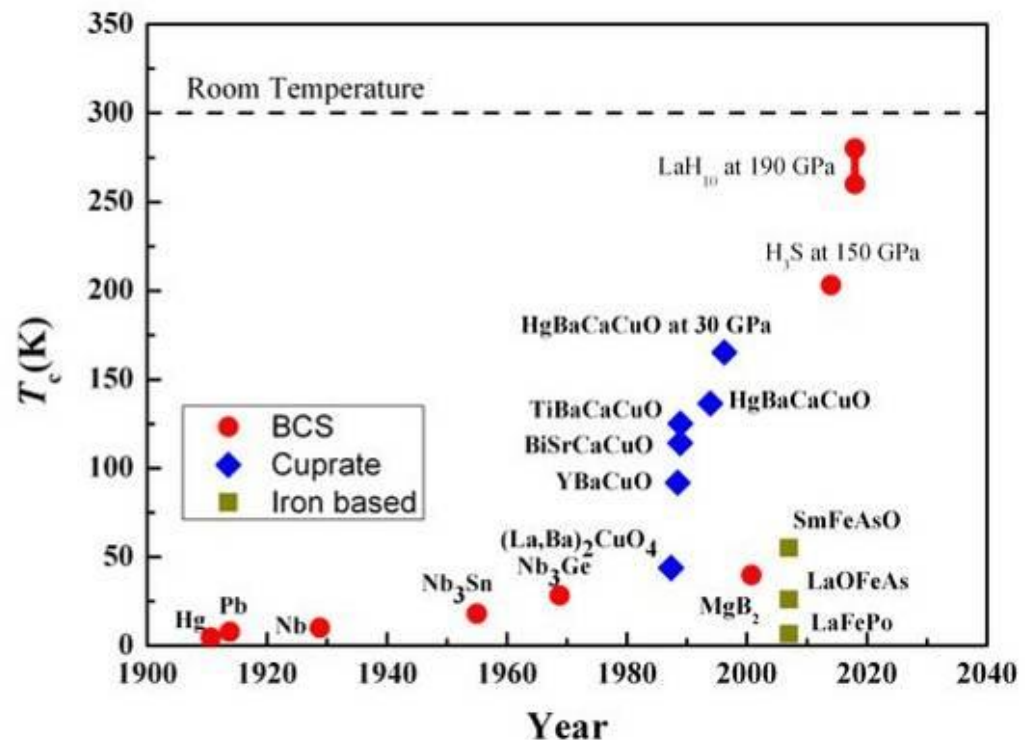
**The Physicist Who
Won the Nobel Prize
Twice!**



**Bardeen, Cooper and Schreffer
1972 Nobel Prize in Physics**

Chasing High T_c

- The BCS theory cannot explain superconductors with $T_c > 40$ K
- Theory for high T_c superconductors is still not complete



Thank you for your attention